Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division

Digital Learning Handout

Part A: Content Design

Course Title	Natural Language Processing Applications
Course No(s)	AIMLCZG519
Credit Units	4
Credit Model	3-1-0
Course Author	Prof. Chetana Gavankar
Version No:	4.0
Date:	8 March 2026

Course Description:

Sentiment Analysis & Opinion Mining, Machine Translation including Indic Languages, Question answering and Conversational AI, Information extraction, Knowledge graph (RAG) Applications, Security, Privacy & Ethics in NLP, Scalable & Efficient NLP Systems, Implementation of above NLP applications using state of art Generative AI and Agentic AI technologies

Course Objectives

No	Course Objective
CO1	To provide students with the knowledge on designing and applying algorithms for real life NLP Applications
CO2	To give an overview of the major technologies used in NLP and hands-on experience of using such tools
CO3	To apply NLP techniques in state of art applications like Machine Translation, Information Extraction including Named entity recognition and Relation extraction
CO4	To give students opportunities to sharpen their programming skills for Natural Language Processing applications

Text Book(s):

T1	Jurafsky and Martin, SPEECH and LANGUAGE PROCESSING: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition, McGraw Hill, Jan 2025

Reference Book(s) & other resources:

R1	Manning and Schütze, Foundations of Statistical Natural Language Processing, MIT Press. Cambridge, MA
R2	Neural Machine Translation by Philipp Koehn

Learning Outcomes: Students will be able to

LO1	To provide students with the knowledge on designing and applying algorithms for real life NLP Applications
LO2	To give an overview of the major technologies used in NLP and hands-on experience of using such tools
LO3	To apply NLP techniques in state of art applications like Machine Translation, Information Extraction including Named entity recognition and Relation extraction
LO4	To give students opportunities to sharpen their programming skills for Natural Language Processing applications

Modular Content Structure

1. **Overview of the course**
2. **Grammar and spellcheckers**
3. **Retrieval Augmented Generation**
   - Naïve RAG, Graph RAG
   - Agentic RAG
   - Practical real world case studies of RAG
4. **Question Answering and Conversational AI**
   - **Question Answering**
   - **Conversational AI**
   - Implementation of these systems using state of art Gen AI techniques including LLM, RAG, Agentic AI techniques
5. **Machine Translation**
   - Techniques of machine translation
   - Indic Language Translation
6. **Information Extraction**
   - Named Entity Recognition
   - Relation Extraction
   - Extracting Events and Time
7. **Sentiment Analysis**
   - Sentiment Analysis Methods
   - Gen AI and Agentic AI approaches for Sentiment Analysis

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Part B: Learning Plan

Contact Session	List of Topic Title	Sub-Topics	Reference
1	**Introduction**	Overview of interesting NLP applications	web resources
2	**Spellcheck and Grammar check**	• Techniques for implementing Automatic Spell checkers and Grammar checkers	Research papers and web resources
3	**Retrieval Augmented Generation (RAG) applications**	• Naïve RAG, Graph RAG • Agentic RAG • Practical real world case studies of RAG	Research papers and web resources
4	**Question Answering**	• Introduction to Agentic AI techniques • Gen AI techniques and Agentic AI use in QA systems	T1- chapter 14 + additional web resources
5	**Conversational AI**	• Hybrid systems • Implementation of Conversational AI systems using state of art Gen AI techniques including LLM, RAG, Agentic AI techniques	T1- chapter 15 + web resources
6	**Sentiment Analysis**	• Sentiment Analysis Methods • Neural Networks for Sentiment Analysis	T1- chapter 4 + web resources
7	**Privacy and Ethics in NLP Applications**	• Data Issues • Model Issues • Deployment Issues	web resources
8	**Premidsem Review**	**Session 1 to Session 7 Review**	
9	**Machine Translation**	• Statistical Approaches	T1- chapter 13
10	**Machine Translation**	• Neural Approaches	T1- chapter 13
11	**Machine Translation**	• Indic Language machine translation	web resources
12	**Information Extraction**	• Named Entity Recognition	T1- chapter 20+ web resources
13	**Information Extraction**	• Relation Extraction	T1- chapter 20+ web resources
14	**Information Extraction**	• Extracting Events and Time	T1- chapter 20+ web resources
15	**Scalable & Efficient NLP Systems**	• Need for Efficiency & Scalable Data • Model Compression & Efficient Architectures • Inference Optimization & Deployment	Research Papers web resources
16	**Postmidsem Review**	**Session 9 to Session 15 Review**	

Experiential Learning Components:

Describe objective, outcome of Experiential Learning Component and the lab infrastructure needed (virtual, remote, open source etc...) number of lab exercises needed, etc.

1. Lab work: 10
2. Project work: 0
3. Case Study: 4 Webinars
4. Simulation: 0
5. Work Integrated Learning Assignment- 2 Assignments with research component
6. Design work/ Field work: 0

Objective of Experiential Learning Component:

Hands on sessions on implementation of fundamental NLP Applications using state of art tools

Scope of Experiential Learning Component:

Programming language - Python

Tools and libraries: SPACY, OpenAI, neo4j, Huggingface etc

Lab Infrastructure:

Virtual Labs / Open source api

List of Experiments:

Lab No.	Lab Objective	Session Reference
1	**Introduction to open source tools used for state of art NLP Applications**	1
2	**Spellcheck and Grammar check**	2
3	**Retrieval Augmented Generation (RAG) applications**	3
4	**Question Answering and Conversational AI**	4, 5
5	**Sentiment Analysis**	6
6	**Privacy and Ethics in NLP Applications**	7
7	**Machine Translation**	9,10,11
8	**Information Extraction: Named entity extraction**	12
9	**Information Extraction: Event Extraction**	13,14
10	**Scalable & Efficient NLP Systems**	15

Evaluation Scheme:

Legend: EC = Evaluation Component; AN = After Noon Session; FN = Fore Noon Session

Evaluation Component	Name (Quiz, Lab, Project, Mid-term exam, End semester exam, etc.)	Type (Open book, Closed book, Online, etc.)	Weigh t	Duration	Day, Date, Session, Time
EC – 1*	Quiz	Online	10%	1 week	To be announced
	Assignment/Lab Assignment / Lab Exams	Online	25%	10 days	To be announced
EC - 2	Mid-Semester Test	Closed Book	30%	2 hours	To be announced
EC - 3	Comprehensive Exam	Open Book	35%	2 ½ Hours	To be announced

EC1* (20% - 30%): Quiz (optional): 5-10 %, Lab Assignment/Assignment: 20% - 30%
Syllabus for Mid-Semester Test (Closed Book): Topics in Contact session: 1 to 8
Syllabus for Comprehensive Exam (Open Book): All topics

Important Links and Information:

eLearn Portal: https://elearn.bits-pilani.ac.in

Students must visit the eLearn portal regularly and stay updated with the latest announcements and deadlines.

Contact Sessions: Students should attend the online lectures as per the schedule provided on the eLearn portal.

Evaluation Guidelines:

1. EC-1 consists of either two Assignments or three Quizzes. Students will attempt them through the course pages on the eLearn portal. Announcements will be made on the portal in a timely manner.
2. For Closed Book tests: No books or reference material of any kind will be permitted.
3. For Open Book exams: "open book" means text/ reference books (publisher copy only) and does not include any other learning material. No other learning material will be permitted during the open book examinations. For Detailed Guidelines refer to the attached document.
   EC3 Guidelines
4. If a student is unable to appear for the Regular Test/Exam due to genuine exigencies, the student should follow the procedure to apply for the Make-Up Test/Exam, which will be made available on the eLearn portal. The Make-Up Test/Exam will be conducted only at selected exam centres on the dates to be announced later.

It shall be the responsibility of the individual student to be regular in maintaining the self-study schedule as given in the course handout, attend the online lectures, and take all the prescribed evaluation components such as Assignments/Quizzes, Mid-Semester Tests and Comprehensive Exams according to the evaluation scheme provided in the handout.

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